

Supplementary Note 1: Sensitivity of survival dynamics to the instability rate α and feedback strength g

Motivation

A reviewer asked: “...a parameter sensitivity analysis would be useful. For example, what happens if only one parameter (α or g) is changed at a time?” The two coefficients enter the Langevin drift asymmetrically, so there is no a priori reason to expect the survival curve to respond to them in the same way. We quantify this directly by perturbing α and g one at a time around the fitted DMSO baseline and comparing the resulting shift in simulated survival.

Model and baseline

Organismal deterioration is modeled as the stochastic process (main text, Eq. 1)

$$\frac{dz}{dt} = \alpha z + gz^2 + \sqrt{2D} \eta(t), \quad (1)$$

integrated by Euler–Maruyama with fixed step $\Delta t = 0.1$ (matching the step used to fit the main-text parameters), $z(0) = 0$, and death defined as the first-passage event $z(t) \geq Z \equiv \alpha/g$. The DMSO baseline used throughout the main text (Table I) is

$$\alpha_0 = 0.155 \text{ d}^{-1}, \quad g_0 = 2.86 \times 10^{-3}, \quad Z_0 = \alpha_0/g_0 = 54.3, \quad D = 1.27, \quad (2)$$

with D fixed at this value throughout, exactly as in the intervention fits: the reviewer’s question concerns α and g specifically, and holding D fixed isolates their effect from the (separately estimated) noise power.

It is useful to track the dimensionless ratio

$$\gamma \equiv \frac{\alpha Z^2}{2D}, \quad (3)$$

which sets the model’s dynamical regime: $\gamma \gg 1$ is the weak-nonlinearity, “Gompertzian” regime in which the population’s average lifespan greatly exceeds the intrinsic instability timescale α^{-1} [1]. All parameter combinations used to fit the main-text conditions satisfy $\gamma \geq 50$.

Sweep design

We perturb the baseline by a common set of multiplicative factors, $\{0.5, 0.75, 1, 1.25, 1.5\}$ (i.e. $\pm 50\%$, $\pm 25\%$, and the baseline itself), in two one-at-a-time sweeps:

- **Sweep A:** vary α , hold $g = g_0$ fixed. Because $Z = \alpha/g$, the failure threshold moves *with* α in this sweep.
- **Sweep B:** vary g , hold $\alpha = \alpha_0$ fixed. Z then moves *inversely* with g .

Substituting $Z = \alpha/g$ into Eq. (3) gives $\gamma = \alpha^3/(2Dg^2)$, so **holding g fixed, γ scales as α^3** , while **holding α fixed, γ scales as $1/g^2$** . A cubic dependence is markedly steeper than an inverse-square one over the same $\pm 50\%$ range, so Sweep A is expected to move the survival curve

considerably more than Sweep B – a quantitative prediction we check directly below rather than merely inspecting the curves visually.

As an outcome metric we report the median simulated lifespan T_{50} (the time at which the population’s own simulated Kaplan–Meier curve crosses $S = 0.5$), rather than the mortality-rate doubling time $\text{MRDT} = \ln 2/\alpha$: MRDT depends on α alone and is, by construction, exactly unchanged across the entirety of Sweep B, even though the survival curve itself visibly shifts (Fig. 1, right panel) – reporting MRDT alone would make Sweep B look inert. T_{50} is obtained from direct stochastic simulation of Eq. (1) (2×10^4 paths per parameter set), not from the closed-form Gompertz survival function, because that closed form is only derived to hold for $\gamma \gg 1$: Sweep A’s $\alpha \times 0.5$ point falls to $\gamma \approx 22.6$, below the $\gamma \geq 50$ bound used to constrain the main-text fits, where the closed-form approximation is no longer guaranteed reliable.

Results

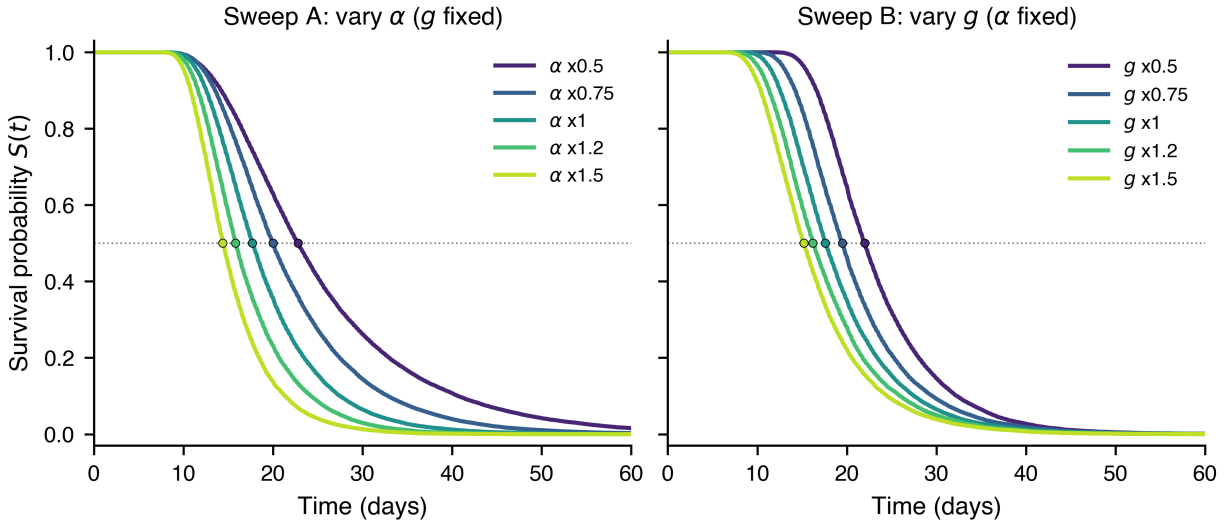


Figure 1: Simulated Kaplan–Meier survival curves under one-at-a-time perturbation of α (left) or g (right), $\pm 50\%$ and $\pm 25\%$ around the DMSO baseline (teal). Circles mark each curve’s median survival time T_{50} . The α sweep visibly spreads the curves more than the equivalent g sweep.

Table 1 confirms the prediction numerically: across the full $\pm 50\%$ range, Sweep A spans $T_{50} \in [14.4, 22.9]$ d (range 8.5 d) while Sweep B spans $T_{50} \in [15.2, 22.0]$ d (range 6.8 d). γ likewise swings far more under Sweep A (22.6 to 609) than under Sweep B (80.2 to 722), consistent with the α^3 vs. $1/g^2$ scaling derived above.

Conclusion

α is the more powerful lever on survival outcome in this model: equal relative perturbations to α shift the median lifespan more than equivalent perturbations to g , a direct, derivable consequence of how the two coefficients enter the dimensionless regime parameter γ once $Z = \alpha/g$ is substituted in. This is not a symmetry one should expect a priori from the drift equation’s literal coefficients,

Sweep	Factor	α (d ⁻¹)	g (10 ⁻³)	Z	γ	MRDT (d)	T_{50} (d)
Sweep A (α varies)	0.5	0.0776	2.86	27.2	22.6	8.93	22.9
Sweep A (α varies)	0.75	0.116	2.86	40.8	76.1	5.95	19.9
Sweep A (α varies)	1	0.155	2.86	54.3	180	4.46	17.6
Sweep A (α varies)	1.25	0.194	2.86	67.9	352	3.57	15.8
Sweep A (α varies)	1.5	0.233	2.86	81.5	609	2.98	14.4
Sweep B (g varies)	0.5	0.155	1.43	109	722	4.46	22
Sweep B (g varies)	0.75	0.155	2.14	72.4	321	4.46	19.4
Sweep B (g varies)	1	0.155	2.86	54.3	180	4.46	17.6
Sweep B (g varies)	1.25	0.155	3.57	43.5	115	4.46	16.3
Sweep B (g varies)	1.5	0.155	4.29	36.2	80.2	4.46	15.2

Table 1: One-at-a-time sensitivity of the DMSO baseline ($\alpha_0 = 0.155$, $g_0 = 2.86 \times 10^{-3}$, $Z_0 = 54.3$, $D = 1.27$). $\gamma = \alpha Z^2 / (2D)$; T_{50} is the median simulated lifespan from 2×10^4 Euler–Maruyama paths of Eq. (1), not the closed-form Gompertz approximation, which requires $\gamma \gg 1$ and is unreliable at Sweep A’s low- α end.

and it is consistent with the fitting procedure’s own emphasis on α as the primary determinant of each condition’s survival curve (main text, Table I).

References

- [1] D. Podolskiy, I. Molodtcov, A. Zenin, V. Kogan, L. I. Menshikov, V. N. Gladyshev, *et al.*, *Critical dynamics of gene networks is a mechanism behind aging and Gompertz law*, arXiv:1502.04307 (2016).